THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL UNIT TEST SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A hemispherical depression of diameter 42 cm, the largest possible, is scooped out of the top face of a cubical block of edge 42 cm. The volume of the remaining block is: 1

(a) 19404 cm³ **(b)** 54684 cm³ **(c)** 35280 cm³ **(d)** 74088 cm³

2. The mean of a distribution with classes 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 13, 11, 2, p, 14, 5 respectively is 37.2. The value of p is: 1

(a) 10 **(b)** 4 **(c)** 5 **(d)** 7

3. A bag contains 4 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 12 **(b)** 16 **(c)** 11 **(d)** 3

4. A solid metallic sphere of radius 9 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. The length of the wire is: 1

(a) 121.5 cm **(b)** 486 cm **(c)** 729 cm **(d)** 243 cm

5. The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80 and frequencies 7, x, 16, y, 3, 4 respectively, and total frequency 50, is 45. The value of x is: 1

(a) 11 **(b)** 12 **(c)** 10 **(d)** 9

6. A bag contains 4 red and 6 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{5}{8}$. The value of x is: 1

(a) 6 **(b)** 5 **(c)** 12 **(d)** 7

7. A solid sphere of radius 6 cm is completely dropped into water in a cylindrical vessel of base radius 12 cm, and no water spills. The rise in the water level is: 1

(a) 0.5 cm **(b)** 1 cm **(c)** 2 cm **(d)** 4 cm

8. The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 7, 14, x, 20, y, 19, 3 respectively, and total frequency 100, is 54. The value of y is: 1

(a) 18 **(b)** 22 **(c)** 16 **(d)** 20

9. A red die and a blue die are rolled together. The probability that at least one die shows 6 is: 1

(a) $\frac{25}{36}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{11}{36}$ **(d)** $\frac{5}{18}$

10. A capsule is a cylinder of radius 14 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 6688 cm² **(b)** 3344 cm² **(c)** 4576 cm² **(d)** 2112 cm²

11. The median class of a distribution is the class whose cumulative frequency is: 1

(a) just greater than $\frac{N}{2}$ **(b)** equal to N always

(c) just less than $\frac{N}{2}$ **(d)** the highest frequency

12. A letter is chosen at random from the word 'PROBABILITY'. The probability that it is a vowel is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{7}{11}$

13. A toy is a cone of base radius 28 cm and height 96 cm mounted on a hemisphere of the same radius (the flat faces coincide). The total surface area of the toy is: 1

(a) 16192 cm² **(b)** 8800 cm² **(c)** 13376 cm² **(d)** 13728 cm²

14. For the data with classes 70–80, 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 with frequencies 8, 19, 19, 20, 16, 9, 9 respectively, The mean of the data is: 1

(a) 103 **(b)** 105 **(c)** 104 **(d)** 102

15. A red die and a blue die are rolled together. The probability that the sum is 12 is: 1

(a) $\frac{1}{36}$ **(b)** $\frac{35}{36}$ **(c)** 0 **(d)** $\frac{1}{18}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	80–90	90–100	100–110	110–120	120–130	130–140
Number of workers	3	6	13	15	7	6

Find the mean.

17. A red die and a blue die are rolled together. Find the probability that one number is exactly double the other. 2

18. A solid metallic cylinder of base radius 6 cm and height 72 cm is melted and recast into solid spheres of radius 2 cm. Find the number of spheres obtained. 2

19. The daily wages (in ₹) of workers in a factory are given below. 2

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60	60–70	70–80
Number of workers	8	2	3	20	3	3	11

Find the median.

20. 42 cards numbered 1 to 42 are placed in a box, mixed well and one is drawn at random. Find the probability that the number drawn is a prime number. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A bag contains 5 red and 5 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{3}$. Find the value of x. 3

22. A wooden article is made by scooping out a hemisphere of radius 7 cm from each end of a solid cylinder of height 10 cm and base radius 7 cm. Find the total surface area of the article. 3

23. The daily wages (in ₹) of workers in a factory are given below. 3

Daily wage (₹)	10–20	20–30	30–40	40–50	50–60	60–70
Number of workers	13	4	6	8	14	5

Find the mean and the mode (in this order).

24. A red die and a blue die are rolled together. Find the probability that the sum of the numbers is 7. 3

OR

A capsule is a cylinder of radius 21 cm and length 12 cm (straight part) with a hemisphere of the same radius fixed at each end. Find the volume of the capsule. 3

25. The ages (in years) of patients admitted to a hospital in a month are given below. 3

Age (years)	10–20	20–30	30–40	40–50	50–60	60–70
Number of patients	5	5	7	6	9	8

Find the mean.

OR

A bag contains 9 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) one number is exactly double the other,
(ii) the product is less than 8,
(iii) the larger number is 4.

OR

Water from a full cylindrical tank of radius 7 cm and height 30 cm is poured into an empty rectangular tank of base 21 cm by 11 cm. Find the height of water in the rectangular tank. 5

27. The median of a distribution with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140, 140–150 and frequencies 3, 8, 16, x, 20, y, 10 respectively, and total frequency 100, is 120.5. Find the value of x. 5

OR

A box contains 50 cards numbered 1 to 50. One card is drawn at random. Find the probability that the number is: 5
(i) a perfect square,
(ii) a number divisible by 5,
(iii) an even prime number (in this order).

END OF QUESTION PAPER — Total: 50 Marks

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